APPROXIMATION BY PERFECT COMPLEXES HOLDS ON AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-affine-compare-bounded is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Representing cohomology classes by dual Koszul complexes. lemma-represent-cohomology-class-on-closed (carded L26768). Produces the maps K_e[-t] -> E realizing sections.
- Finite type of top cohomology. Cohomology, lemma finite-cohomology; Properties, lemma finite-type-module (cross-chapter, named). H^t(E) is of finite type, generated by finitely many sections.
- Cohomology of pseudo-coherent complexes on affines. lemma-affine-compare-bounded (same chapter, NOT carded, declared external). Identifies Gamma(X, H^t(E)) with H^t(X, E).
- Pseudo-coherence of cones. Cohomology, lemma cone-pseudo-coherent (cross-chapter, named). E' stays m-pseudo-coherent in the triangle.
- Two-out-of-three for perfect complexes. Cohomology, lemma two-out-of-three-perfect (cross-chapter, named). P" is perfect in the assembled triangle.

Source reference. Original reference 08CD: focused-seeds/geometry/S9-perf2-approximation-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DN: focused-seeds/geometry/S9-perf2-approximation-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08CR: focused-seeds/geometry/S9-perf2-approximation-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 08EM: sources/stacks-perfect/pinned-original-source.tex, source lines 3814–3814. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E3: sources/stacks-perfect/pinned-original-source.tex, source lines 2374–2374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PB: focused-seeds/geometry/S9-perf2-approximation-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be an affine scheme. Then approximation holds for every triple (T, E, m) as in Definition 08EM such that there exists an integer $r \geq 0$ with*

(1) *E is m-pseudo-coherent,*
(2) *$H^i(E)$ is supported on T for $i \geq m - r + 1$,*
(3) *$X \setminus T$ is the union of r affine opens.*

In particular, approximation by perfect complexes holds for affine schemes.

Proof. Say $X = \mathrm{Spec}(A)$. Write $T = V(f_1, \ldots, f_r)$. (The case $r = 0$, i.e., $T = X$ follows immediately from Lemma 08E7 and the definitions.) Let (T, E, m) be a triple as in the lemma. Let t be the largest integer such that $H^t(E)$ is nonzero. We will proceed by induction on t. The base case is $t < m$; in this case the result is trivial. Now suppose that $t \geq m$. By Cohomology, Lemma 08DN the sheaf $H^t(E)$ is of finite type. Since it is quasi-coherent it is generated by finitely many sections (Properties, Lemma 01PB). For every $s \in \Gamma(X, H^t(E)) = H^t(X, E)$ (see proof of Lemma 06Z0) we can find an $e > 0$ and a morphism $K_e[-t] \to E$ such that s is in the image of $H^0(K_e) = H^t(K_e[-t]) \to H^t(E)$, see Lemma 08E3. Taking a finite direct sum of these maps we obtain a map $P \to E$ where P is a perfect complex supported on T, where $H^i(P) = 0$ for $i > t$, and where $H^t(P) \to E$ is surjective. Choose a distinguished triangle

$$P \to E \to E' \to P[1]$$

Then E' is m-pseudo-coherent (Cohomology, Lemma 08CD), $H^i(E') = 0$ for $i \geq t$, and $H^i(E')$ is supported on T for $i \geq m - r + 1$. By induction we find an approximation $P' \to E'$ of (T, E', m). Fit the composition $P' \to E' \to P[1]$ into a

distinguished triangle $P \to P'' \to P' \to P[1]$ and extend the morphisms $P' \to E'$ and $P[1] \to P[1]$ into a morphism of distinguished triangles

$$\begin{array}{ccccccc} P & \longrightarrow & P'' & \longrightarrow & P' & \longrightarrow & P[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ P & \longrightarrow & E & \longrightarrow & E' & \longrightarrow & P[1] \end{array}$$

using TR3. Then P'' is a perfect complex (Cohomology, Lemma 08CR) supported on T. An easy diagram chase shows that $P'' \to E$ is the desired approximation. $\square$